

# Government Budget Analysis & Visualization Dashboard

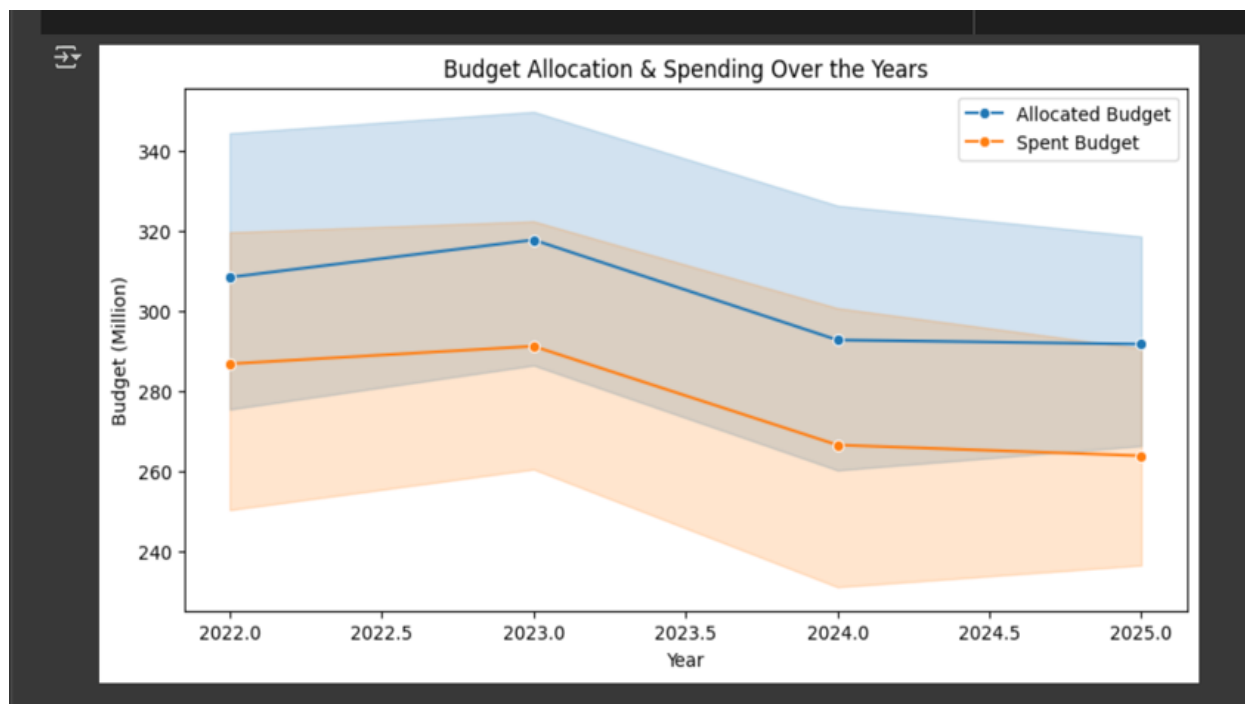
## Project Overview

This project focuses on analyzing government public spending using a data-driven dashboard. The dashboard visualizes budget allocations, expenditures, and key financial indicators, helping policymakers make informed decisions about resource distribution and efficiency.

## Key Visualizations & Insights

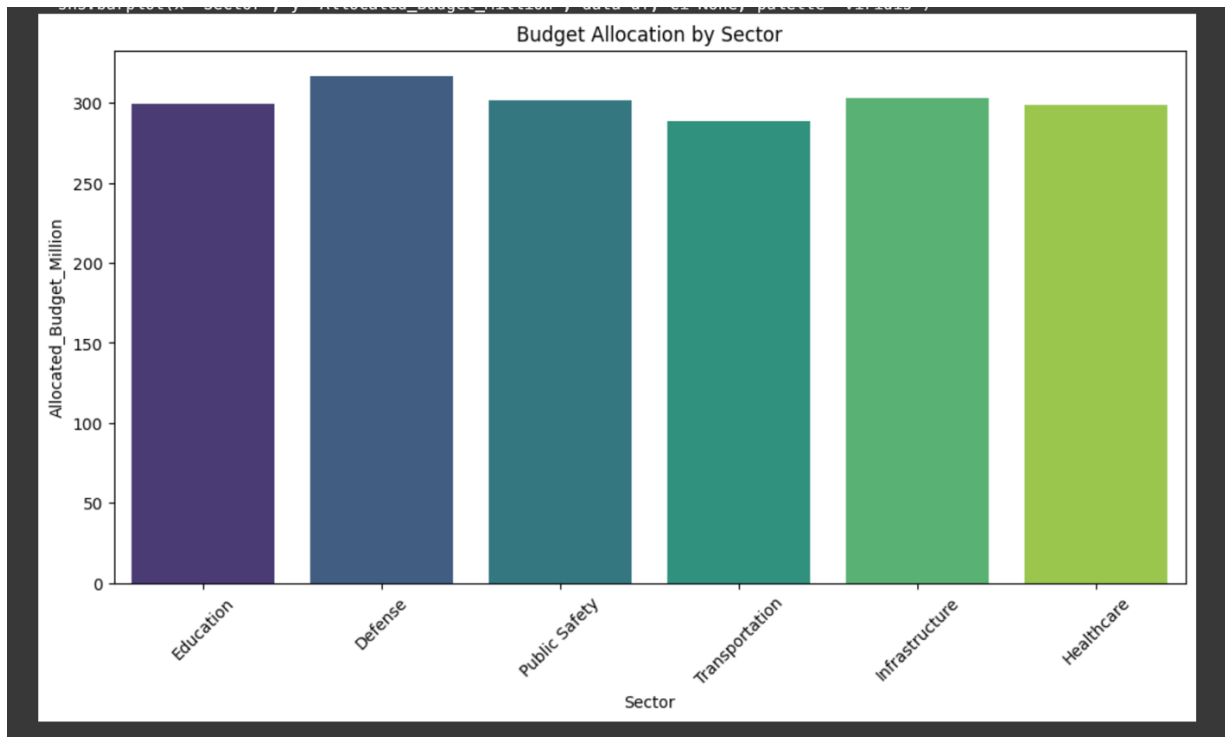
### 1. Budget Allocation & Spending Trends Over the Years

This line chart illustrates the trends of allocated, spent, and projected budgets from 2022 to 2025. The shaded region highlights the gap between allocated and spent budgets, offering insights into budget efficiency over time.



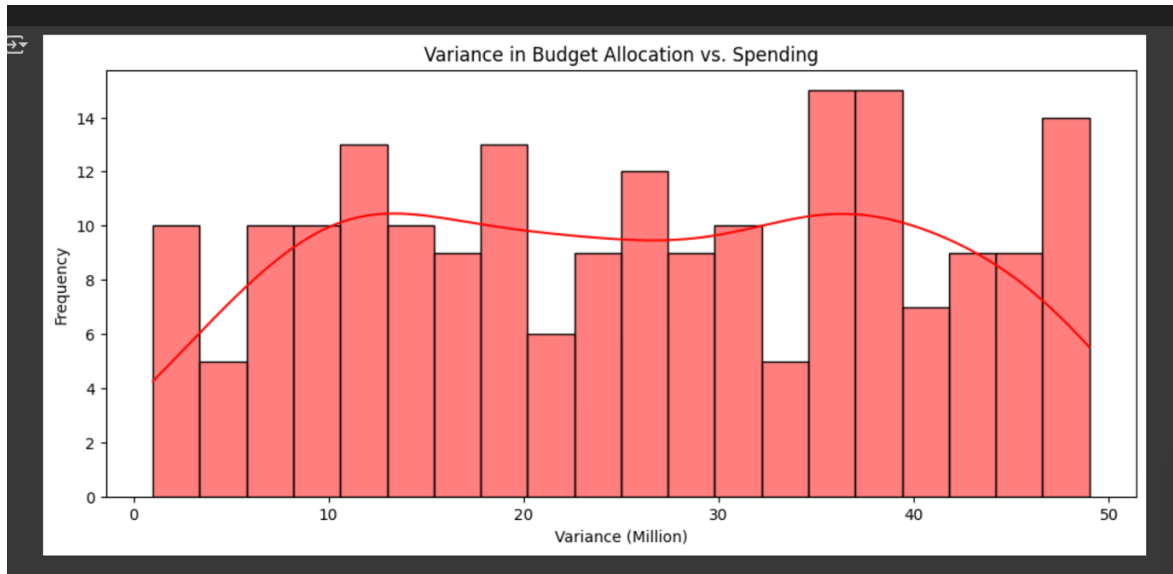
## 2. Budget Allocation by Sector

A bar chart showing how the budget is distributed across major sectors like Education, Defense, Healthcare, Infrastructure, and Transportation. This visualization helps understand financial priorities and funding allocations.



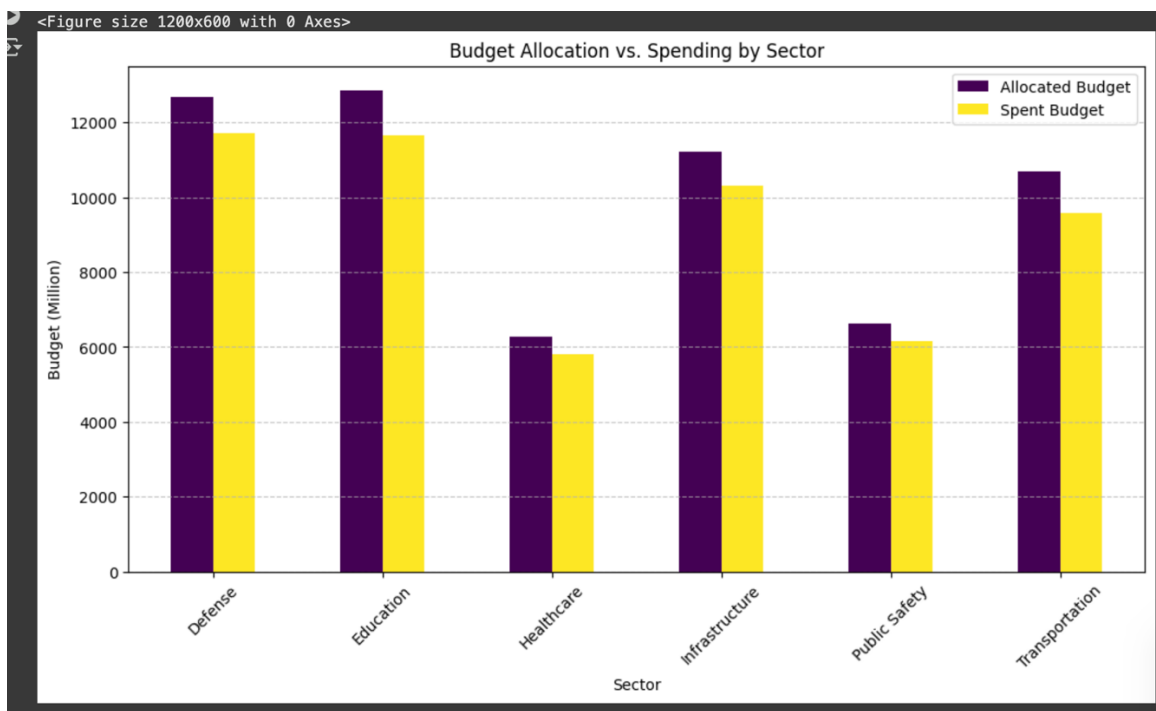
## 3. Variance in Budget Allocation vs. Spending

A histogram with KDE (Kernel Density Estimation) that visualizes the frequency of budget variances, revealing how often government spending deviates from its allocated budget. A higher variance suggests inefficiencies or unexpected costs.



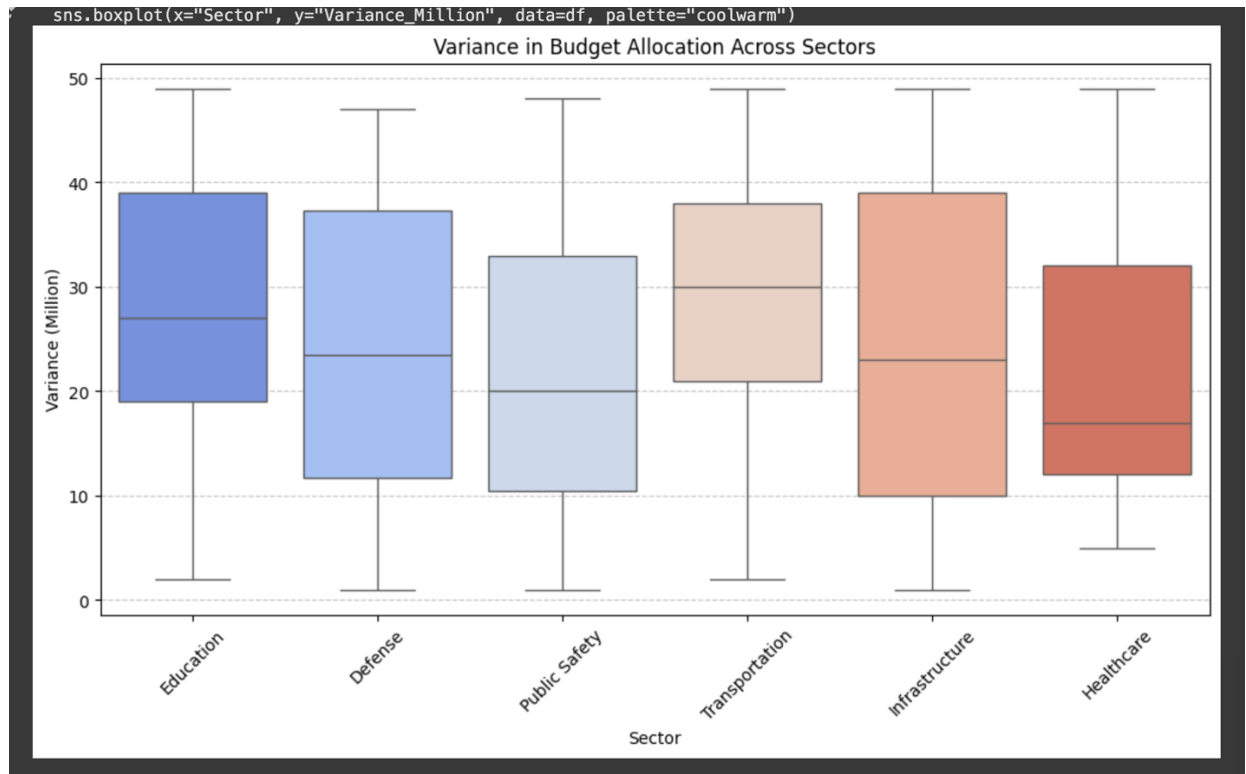
#### 4. Budget Allocation vs. Spending by Sector

This grouped bar chart compares the allocated budget vs. actual spending for different sectors. It highlights underfunded and overfunded sectors, offering insights into potential areas for financial adjustment.



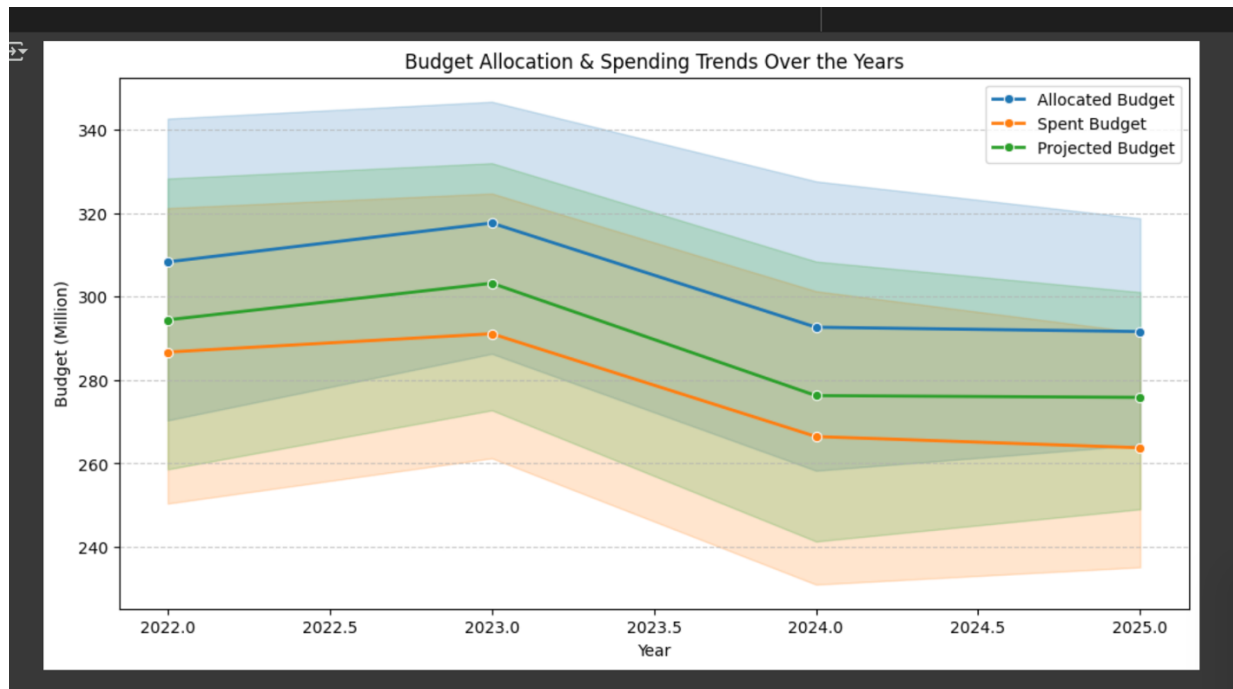
## 5. Variance in Budget Allocation Across Sectors

A box plot displaying the spread of budget variances across sectors. This chart helps identify which sectors consistently receive excess or inadequate funding, aiding in better budget planning.



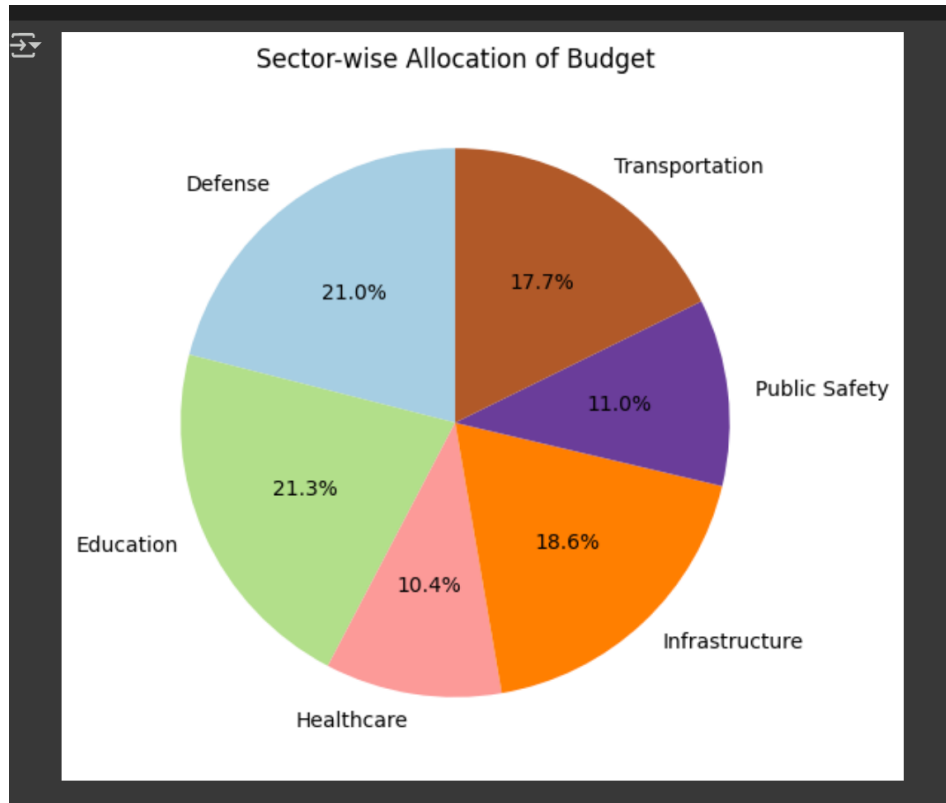
## 6. Budget Allocation & Spending Trends Over the Years (with Projection)

A more detailed line chart, integrating future budget projections alongside allocated and spent budgets. This visualization helps policymakers assess whether past spending trends align with future financial planning.



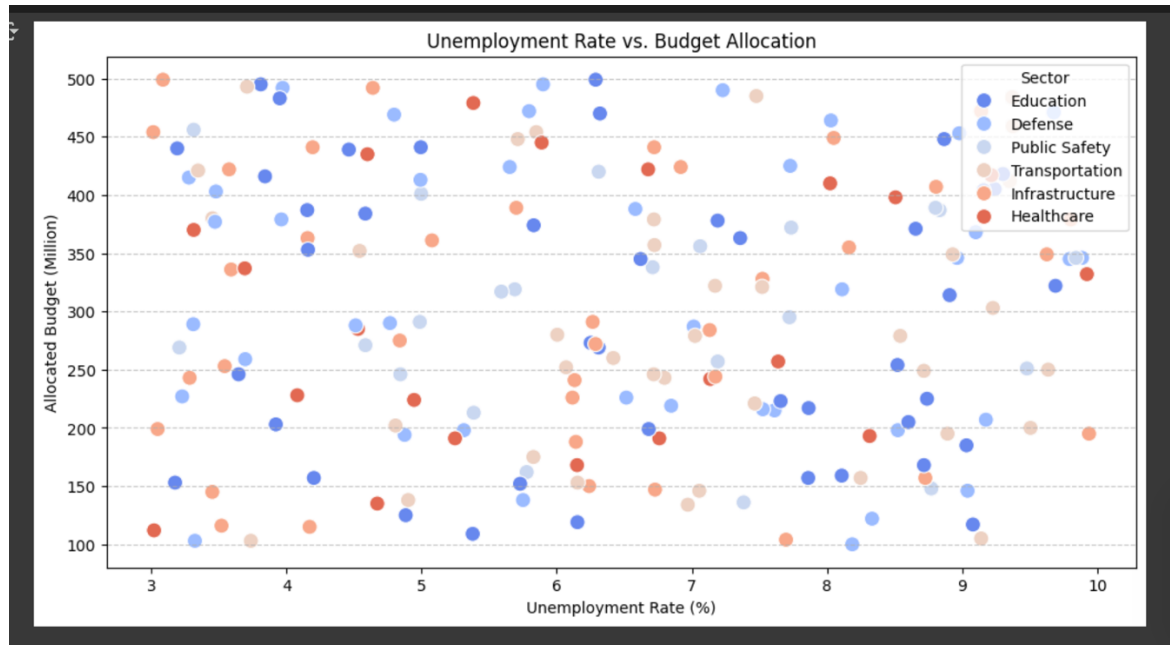
## 7. Sector-wise Allocation of Budget

A pie chart illustrating the percentage of government spending for each sector. This provides a quick snapshot of fund distribution, helping analysts evaluate sectoral spending efficiency.



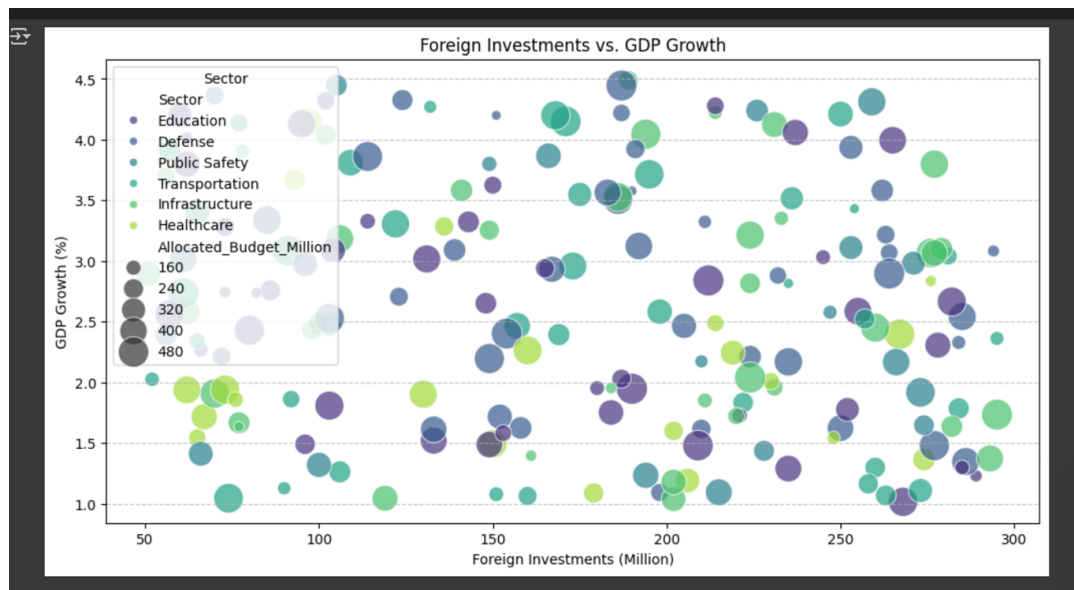
## 8. Unemployment Rate vs. Budget Allocation

A scatter plot showcasing the relationship between budget allocation and unemployment rates across various sectors. This helps in analyzing whether investments in specific sectors lead to reduced unemployment.



## 9. Foreign Investments vs. GDP Growth

A bubble chart displaying the correlation between foreign investments and GDP growth across multiple sectors. The size of the bubbles represents the allocated budget, helping identify which sectors contribute the most to economic growth.



## 10. Public Opinion Score vs. Social Welfare Spending

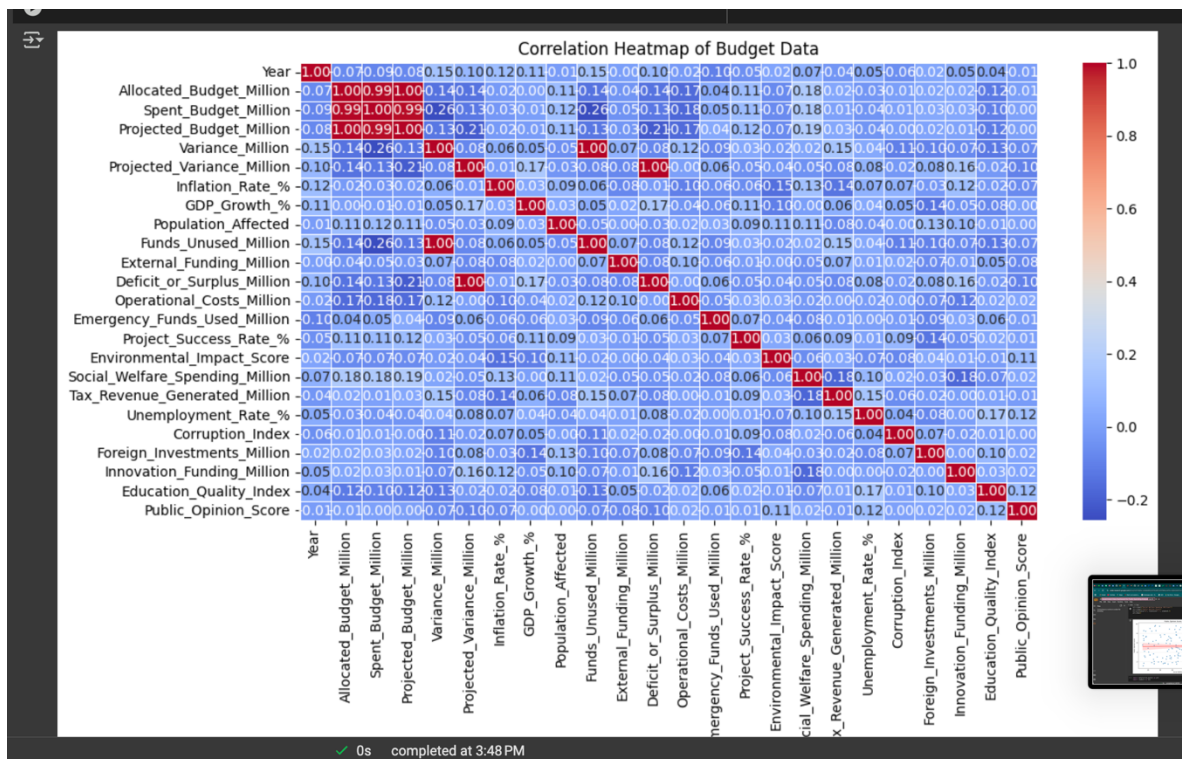
A scatter plot with regression trendline that analyzes the impact of social welfare spending on public opinion scores. This visualization helps in evaluating whether higher social investments lead to better public satisfaction and trust in governance.



## 11. Correlation Heatmap of Budget Data

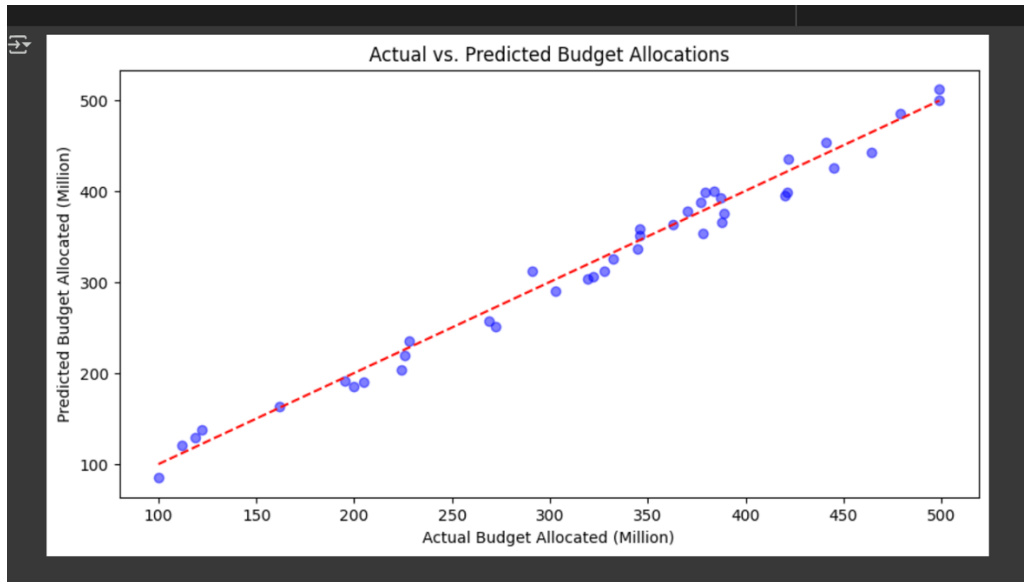
A correlation heatmap visualizing the relationships between various budget metrics, economic indicators, and public sentiment. It helps identify key influencing factors for budget planning.





## 12. Actual vs. Predicted Budget Allocations

A scatter plot with a regression line, showing the performance of a predictive ML model used to estimate future budget allocations. The model achieved a reasonable  $R^2$  score, indicating its accuracy in budget forecasting.



### Final Insights:

- ✓ Identified budget spending trends across different years & sectors.
- ✓ Analyzed key financial indicators like GDP growth, inflation, and tax revenue.
- ✓ Developed a predictive ML model to estimate future budget allocations.
- ✓ Model achieved a reasonable performance with  $R^2$  Score for budget prediction.